

The Zoetop: a kinesthetic-aware zoetrope

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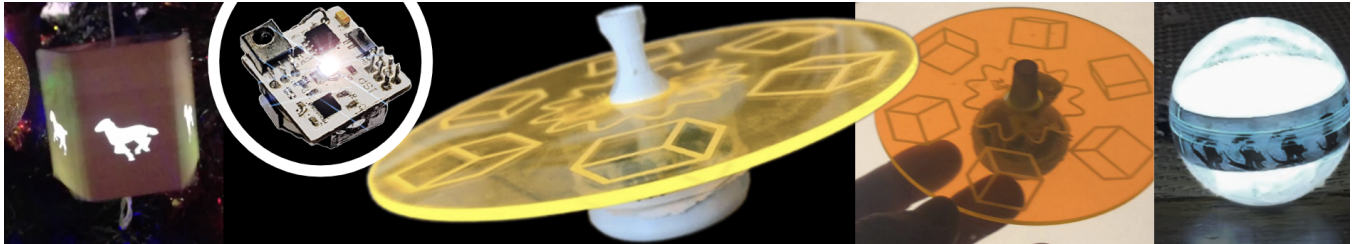


Figure 1: Embedding a gyroscope-controlled miniature strobe on a laser-etched acrylic top produces a Zoetop.

ABSTRACT

We introduce the Zoetop, an innovative variation of the Zoetrope opening up new opportunities for design. At the core of the Zoetop is a small and inexpensive electronic device that can be embedded on any naturally spinning object, such as hand-spun spinning tops, car wheels, bouncing balls, or windmills. The device uses an inertial sensor to keep track of its own instantaneous axis of rotation and angular speed. It then produces a constant number of flashes per revolution regardless of speed variations, and corresponding to the number of frames in the animation. As a consequence animations can be created without strobe light synchronization, delicate mechanical setups including driving motors, or any form of calibration. As such, this “kinaesthetic-aware” computational Zoetrope offers new opportunities for deployment in the wild, augmenting hand-spun toys, wind-powered chimes, propellers or fans with animated graphics. We believe that the Zoetop can inspire designers and the DIY community to come up with new ideas centered around this age-old yet still fascinating Victorian-era optical gadget.

CCS CONCEPTS

• **Hardware** → **Displays and imagers**; • **Computing methodologies** → **Animation**.

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1 INTRODUCTION

As an optical gadget demonstrating the fundamental principle behind every device creating the illusion of a moving image, the zoetrope was never lost to history but evolved alongside analog and digital animation technologies, adopting its functional mechanisms including driving motors and frame synchronisation hardware. In this work, we propose a technological update suited for the zoetrope of the XXI century, integrating inertial motion sensors and computational capabilities. A zoetrope capable of measuring its own angular speed can rotate at a variable speed, while maintaining the animation unaffected by adjusting the frequency of the strobing light in real time. In short, our “kinesthetic enhanced” Zoetrope offers new possibilities for design and deployment in the wild by harnessing the natural motion of its support. This comes with two fundamental advantages: (1) no need of a mechanical setup to maintain a fixed axis or rotation and (2) no need of motors and rotary encoders to maintain a constant speed of rotation. This means that anything that can spin can be used as its support, even as its angular velocity and axis of rotation keeps changing or reorienting in space (e.g. a top spun by hand that slows down while its axis keeps precessing, the tires of car regardless of acceleration, a windmill under variable wind, etc.)

2 RELATED WORK

As an optical toy demonstrating the fundamental principle behind every device conceived to produce moving images, the zoetrope appears and reappears in various settings and situations. It is used in kinetic art, op art, as a teaching toy, and is an ideal project to demonstrate the use of control systems in STEM and physical computing. Zoetropes come in various form factors, ranging from small flip-books to large linear zoetropes found in subways. Advancements in lighting and mechanical motion technology have given rise to numerous variations. Although the original zoetrope was powered by hand, the advent of electricity enabled the development of zoetropes using stroboscopic lights synchronized with driving motors. More recently, zoetropes conquered the third dimension thanks to digital fabrication techniques [Lee and Seong 2009]. Fascinating morphing

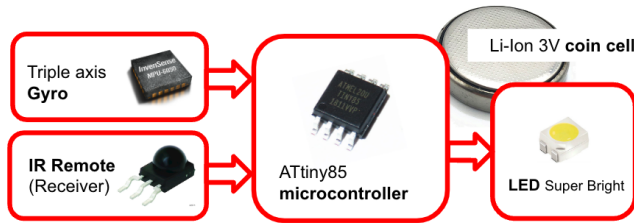


Figure 2: Diagram and main component of the circuit. The user sets the number of frames in the sequence using an IR remote. The gyro measures the average angular velocity and the micro-controller computes the average RPM and the corresponding flashing period

fractals can be printed and subsequently animated using a turntable and synchronized strobing lights, or the animation be seen indirectly on a camera or smartphone by taking advantage of the limited framerate the recorder [Edmark 2017]. Smoot et al. [Smoot et al. 2010] employ aperiodic stroboscopic lights to selectively display "frames" to add interactivity. Miyashita et al. [Miyashita et al. 2016] control the duration of exposure and are so able create the illusion of composite objects.

3 WORKING PRINCIPLE

The core of the Zoetop is a module capable of measuring the angular velocity of its embedding, spinning support using a MEMS gyroscope. From this value and the knowledge of the number of frames in the animation (a value that can be changed at any time by the user using an IR remote), the device estimates the adequate period of the strobe light - a quantity that varies as the spinning object slows down or speeds up. As a consequence, a sharp animation takes place simultaneously in all the sites of the exposed frames, exactly as in the traditional Zoetrope.

Given an animation containing N frames and a given number of revolutions per second (RPS), the number of frames per second (FPS) at which we experience the illusion of the moving image, also corresponding to the flashing frequency, is just: $FPS = N \times RPS$. Using revolutions per minute instead gives $FPS = N \times RPM/60$. If we assume that 12fps is the lower limit for creating the illusion of motion (a number typical in hand-drawn animations, but 24 or 30 produces a smoother effect, and 60fps or more is perfect for all purposes), it follows that the product $N \times RPM$ should always be larger than 720. If we prepare a sequence with N frames, let's say 12, it follows that we will need RPM to be larger than $720/12$, that is 60RPM. This is just one revolution per second, which is not uncommon in everyday life, for instance a toy windmill spinning in light breeze. Conversely, if the RPM is fixed, for instance a rotating ceiling fan at 300RPM, then we have a lower limit on the number of frames equal to $720/300$, that is about 3 frames. An animation with a lower number of frames will work if we rotate faster; however, in the present implementation there is a limit pertaining to the sensing range of the MPU6050 gyroscope (max 330RPM).

Right after being thrown by finger twirling, a spinning top rotates much faster than that. The low-cost gyro employed in our prototype fails to provide adequate measurements at such speed, resulting in

a blurred animation. The case of the spinning acrylic top shown in Figure 1 (see also supplemental video) is illustrative of this non-fundamental, implementation-related limitation. However, as the top slows down, the animation appears visible for a while until the RPM is too low for producing 12fps using just 6 frames in the sequence, in which case the blinking is made to stop to avoid unpleasant low-frequency flashes.

4 PROTOTYPE

The prototype fits on a 25x25mm PCB and consists of an ATtiny85 microcontroller, an IMU (Invensense MPU-6050), an IrDA receiver (TSOP38238), a single omnidirectional super-bright LED (Superflux, 3V/100mA), and is powered by a rechargeable coin battery (Figure 1). The implementation is open source (<https://github.com/daan/zoetop>). Given its size and cost, the device opens up many possibilities for inventive DIY projects, such as paper windmills or spinning tops with printed/etched animations. By simply attaching this module to them, ceiling fans, large wind turbines, or car tires can be turned into zoetrope platforms.

5 FUTURE WORK

A plethora of interesting variations around the Zoetop are possible. One could for instance print multiple animations on the same support; as long as they contain a different number of frames, one animation can be made visible while the other(s) remain blurred or appear to move sideways. More intriguing is the possibility of using sequences composed of colored or dithered images as these could be made to overlap completely, producing the impression of a randomly patterned surface when the support is at rest. Sequence-selectivity can also be exploited to create a human-readable tachometer by using sequences with repeated frames (e.g. numbers or text reading "slow", "medium", or "fast" for instance) or to create composite animations by hopping between sequence's frames. If we limited ourselves to the description of a Zoetop capable of measuring the angular velocity along a *single* axis, it is only for the purpose of clarity. Indeed, the same principle applies for any arbitrary axis of rotation. Any animation lying on a disk perpendicular to this axis will be "activated" while the others will remain blurred. The principle can also be applied to 3D-Zoetropes, which can result in intricate three-dimensional objects that will reveal hidden 3d animations when throw up and spinning in the air, or while rolling on a surface that forces them to alternate between spinning axes as Steinmetz solids do.

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